

A Haar-Rearrangement Solution of the Multiplicity-Two Problem for Multiplication on $L^p[0, 1]$

Candidate solution packet for arXiv:1707.04798, Problem 1

August 9, 2026

Status. Candidate full solution; verdict **likely valid**, pending independent human review. The proof gives an affirmative answer for every $1 < p < \infty$, including $p = 2$.

1 Source question

Let M be multiplication by the coordinate on $L^p[0, 1]$,

$$(Mf)(t) = tf(t).$$

Problem 1 in Section 4 of Boedihardjo's paper [1] asks, for $1 < p < \infty$ and $p \neq 2$, whether there is a compact operator K on $L^p[0, 1]$ such that $M \oplus M$ is similar to $M + K$.

supp(μ_2) \setminus supp(μ'_2). So D_1 and D_2 are similar. Therefore M_{μ_1} and M_{μ_2} are approximately similar. $\square$

4. OPEN PROBLEMS

Problem 1. Let $1 < p < \infty$, $p \neq 2$. Let μ be the Lebesgue measure on $[0, 1]$. Does there exist a compact operator K on $L^p(\mu)$ such that $M_\mu \oplus M_\mu$ is similar to $M_\mu + K$?

Figure 1: Problem 1 on PDF page 9 of the source paper.

The source notes on the following page that the answer is known to be affirmative when $p = 2$.

2 Proof intuition

The coordinate multiplier is compactly close to a diagonal operator on the standard Haar basis: attach to the Haar function on a dyadic interval I the center c_I of that interval. To fit two copies of this diagonal into one copy, send the Haar coordinate indexed by I in the first copy to the coordinate indexed by the left child $I0$, and send the corresponding coordinate in the second copy to the right child $I1$.

There are two points to prove. First, each fixed-child rearrangement is an isomorphism onto its Haar-coordinate range. The upper estimate follows pointwise from Haar square functions; the

lower estimate follows from Stein's vector-valued conditional-expectation inequality. Second, after the rearrangement, the value c_I is placed on $I0$ and $I1$, whereas the original diagonal places c_{I0} and c_{I1} there. The errors have magnitude $|I|/4$, hence tend uniformly to zero by Haar level and define a compact diagonal operator.

3 Dyadic preliminaries

Let $\mathcal{D}$ be the dyadic intervals in $[0, 1]$. For $I \in \mathcal{D}$, let $I0$ and $I1$ denote its left and right children, let c_I be its center, and use the L^p -normalized Haar function

$$h_I = |I|^{-1/p}(\mathbf{1}_{I0} - \mathbf{1}_{I1}).$$

Together with $e = \mathbf{1}_{[0,1]}$, the family $\{e\} \cup \{h_I : I \in \mathcal{D}\}$ is an unconditional Schauder basis of $L^p[0, 1]$.

For a finite Haar expansion $x = \sum_I a_I h_I$, set

$$Sx(t) = \left(\sum_I |a_I|^2 |I|^{-2/p} \mathbf{1}_I(t) \right)^{1/2}.$$

The martingale square-function theorem [2] gives constants depending only on p such that

$$A_p \|x\|_p \leq \|Sx\|_p \leq B_p \|x\|_p. \quad (1)$$

We also use Stein's vector-valued inequality for dyadic conditional expectations [2]: if $\mathbb{E}_n$ is conditional expectation onto the dyadic sigma-algebra at level n , then

$$\left\| \left(\sum_n |\mathbb{E}_n f_n|^2 \right)^{1/2} \right\|_p \leq C_p \left\| \left(\sum_n |f_n|^2 \right)^{1/2} \right\|_p. \quad (2)$$

4 The child rearrangements

Let H be the closed span of the Haar functions h_I .

Lemma 1. *For $\varepsilon \in \{0, 1\}$, the rule*

$$T_\varepsilon h_I = h_{I\varepsilon}$$

extends to an isomorphism from H onto the closed Haar-coordinate subspace

$$Y_\varepsilon = \overline{\text{span}}\{h_{I\varepsilon} : I \in \mathcal{D}\}.$$

Proof. We prove the claim first on finite Haar sums. If $x = \sum_I a_I h_I$, then

$$S(T_\varepsilon x)(t)^2 = 2^{2/p} \sum_I |a_I|^2 |I|^{-2/p} \mathbf{1}_{I\varepsilon}(t) \leq 2^{2/p} Sx(t)^2.$$

Together with (1), this proves the upper bound for T_ε .

For the reverse bound, for every level n define

$$f_n = \sum_{\substack{I \in \mathcal{D} \\ |I|=2^{-n}}} a_I |I|^{-1/p} \mathbf{1}_{I\varepsilon}.$$

The intervals at a fixed level are disjoint, and

$$\mathbb{E}_n f_n = \frac{1}{2} \sum_{|I|=2^{-n}} a_I |I|^{-1/p} \mathbf{1}_I.$$

Consequently,

$$Sx = 2 \left(\sum_n |\mathbb{E}_n f_n|^2 \right)^{1/2}, \quad \left(\sum_n |f_n|^2 \right)^{1/2} = 2^{-1/p} S(T_\varepsilon x).$$

Applying (2), followed by (1) twice, gives

$$\|x\|_p \leq C'_p \|T_\varepsilon x\|_p.$$

Thus T_ε is bounded and bounded below. Its range is therefore closed and is exactly Y_ε . $\square$

Define $U : (L^p[0, 1] \oplus L^p[0, 1])_p \rightarrow L^p[0, 1]$ by

$$U(ae + x, be + y) = ae + bh_{[0,1]} + T_0 x + T_1 y, \quad x, y \in H. \quad (3)$$

The four coordinate families in (3) are disjoint, and together they exhaust the target Haar basis: the constant e , the root Haar function $h_{[0,1]}$, all left-child Haar functions, and all right-child Haar functions. Since the Haar basis is unconditional, the coordinate projections onto these four families are bounded. Lemma 1 therefore shows that U is bounded, onto, and has a bounded inverse.

5 A compact Haar diagonalization

Define a diagonal operator D on the Haar basis by

$$De = \frac{1}{2}e, \quad Dh_I = c_I h_I \quad (I \in \mathcal{D}). \quad (4)$$

Lemma 2. $C = M - D$ is compact on $L^p[0, 1]$.

Proof. Let P_n be the Haar-coordinate projection onto the functions h_I with $|I| = 2^{-n}$. These projections are uniformly bounded by unconditionality. If $x = \sum_{|I|=2^{-n}} a_I h_I$, the supports of the summands are disjoint and $|t - c_I| \leq |I|/2$ on I . Hence

$$\|(M - D)x\|_p \leq 2^{-n-1} \|x\|_p.$$

Let Q_N be the coordinate projection onto the constant and the Haar levels at most N . It follows that

$$\|(M - D)(I - Q_N)\| \leq C_p \sum_{n>N} 2^{-n-1} \longrightarrow 0.$$

Since $(M - D)Q_N$ has finite rank, $M - D$ is the operator-norm limit of finite-rank operators and is compact. $\square$

6 Full solution

Theorem 1. *Let $1 < p < \infty$ and let M be multiplication by t on $L^p[0, 1]$. There are an isomorphism*

$$U : (L^p[0, 1] \oplus L^p[0, 1])_p \longrightarrow L^p[0, 1]$$

and a compact operator K on $L^p[0, 1]$ such that

$$U(M \oplus M)U^{-1} = M + K.$$

In particular, Problem 1 of [1] has an affirmative answer.

Proof. Use the isomorphism U from (3) and the diagonal D from (4). Put

$$A = U(D \oplus D)U^{-1}.$$

This is diagonal in the target Haar basis. The two constant coordinates give

$$Ae = \frac{1}{2}e, \quad Ah_{[0,1]} = \frac{1}{2}h_{[0,1]},$$

and for every dyadic I and $\varepsilon \in \{0, 1\}$,

$$Ah_{I\varepsilon} = c_I h_{I\varepsilon}.$$

On the other hand, $Dh_{I\varepsilon} = c_{I\varepsilon} h_{I\varepsilon}$, and

$$|c_I - c_{I\varepsilon}| = \frac{|I|}{4}.$$

Thus $E = A - D$ is diagonal on an unconditional basis and its diagonal coefficients tend to zero. Its truncations converge in operator norm, so E is compact.

By Lemma 2, $M = D + C$ for a compact C . Therefore

$$\begin{aligned} U(M \oplus M)U^{-1} &= U(D \oplus D)U^{-1} + U(C \oplus C)U^{-1} \\ &= D + E + U(C \oplus C)U^{-1} \\ &= M + (E + U(C \oplus C)U^{-1} - C). \end{aligned}$$

The operator in parentheses is compact. Taking it to be K proves the claim. □

7 Verification notes

The main possible failure modes were checked as follows.

1. The proof does not infer compactness merely from small errors on basis vectors. Lemma 2 gives a summable operator-norm tail, and E is a null diagonal on an unconditional basis.
2. The map $I \mapsto I\varepsilon$ does alter nesting, so arbitrary-basis permutability cannot be invoked. Lemma 1 proves the two required rearrangements directly.
3. The two image subspaces need not be an isometric ℓ^p sum. Bounded coordinate projections give the needed topological direct sum, and finite product norms are equivalent.

4. The finite coordinates match: the first constant maps to the target constant, the second constant to the root Haar function, and all remaining target Haar functions are uniquely a left or right child.

The script `code/verify_dyadic_absorption.py` checks finite dyadic truncations, exact coordinate coverage, parent-child center errors, and random L^p norm ratios. These computations are sanity checks and are not used as proof. With 500 fixed-seed samples at every depth through 8, the exact checks passed and the randomized ratios, over $p \in \{1.2, 1.5, 3, 5\}$, lay between 0.644458 and 1.687973.

8 Novelty and limitations

The lightweight run indexes were searched using arXiv:1707.04798, the exact problem language, multiplicity-two terminology, compact perturbation, and Haar-rearrangement phrases. A bounded external search used the exact question, the paper title and author, the arXiv id, and close variants. It found the source arXiv and journal records but no later work claiming this resolution. The search was bounded rather than exhaustive, so the mathematical novelty claim has moderate confidence.

The theorem resolves Problem 1 and supplies an explicit Haar-coordinate similarity. It does not address Problem 2 in the source paper, which asks for a 1-summing perturbation for a multiplication operator on a square in the complex plane.

9 Human-review recommendation

The packet should be reviewed as a candidate full solution. The highest-value checks are the normalization and applicability of Stein’s inequality in Lemma 1, followed by the finite-coordinate bookkeeping in (3). The compactness and conjugation steps are then routine.

References

- [1] M. T. Boedihardjo, *Multiplication operators on L^p* , Studia Math. 244 (2019), 309–319; arXiv:1707.04798.
- [2] E. M. Stein, *Topics in Harmonic Analysis Related to the Littlewood–Paley Theory*, Annals of Mathematics Studies 63, Princeton University Press, 1970.